Kedro instalation

[The kedro-mlflow plugin — kedro-mlflow 2.0.2 documentation]

Check if your Kedro is installed. If not use the command line to install it in your virtual environment:

```
uv pip install kedro-mlflow
```

To check the installation, write in the console of the terminal:

```
kedro info
```

```
 _            _
| | _____  __| |_ __ ___
| |/ / _ \/ _` | '__/ _ \
|   <  __/ (_| | | | (_) |
|_|\_\___|\__,_|_|  \___/
v1.3.1

Kedro is a Python framework for
creating reproducible, maintainable
and modular data science code.

Installed plugins:
kedro_mlflow: 2.0.2 (entry points:hooks,project)
kedro_telemetry: 0.7.0 (entry points:cli_hooks,hooks)
[05/04/26 23:33:48] INFO     Kedro is sending anonymous usage data with the sole purpose of improving the product. No  plugin.py:242
                             personal data or IP addresses are stored on our side. To opt out, set the
                             `KEDRO_DISABLE_TELEMETRY` or `DO_NOT_TRACK` environment variables, or create a
                             `.telemetry` file in the current working directory with the contents `consent: false`. To
                             hide this message, explicitly grant or deny consent. Read more at
                             https://docs.kedro.org/en/stable/about/telemetry/
```

Now proceed with the configuration of a new project. In the command line write (it will as ask a name for the project later)

```
kedro new
```

```
(test_env) PS C:\Users\rosan\OneDrive\Desktop\trimester_2024\02_modular_code> kedro new

Project Name
============
Please enter a name for your new project.
Spaces, hyphens, and underscores are allowed.
To skip this step in future use --name
 (New Kedro Project): mlops_example
```

It will ask about the extra tools that you want to install. For the moment choose from 1-5,

```
=============
These optional tools can help you apply software engineering best practices.
To skip this step in future use --tools
To find out more: https://docs.kedro.org/en/stable/starters/new_project_tools.html

Tools
1) Lint: Basic linting with Ruff
2) Test: Basic testing with pytest
3) Log: Additional, environment-specific logging options
4) Docs: A Sphinx documentation setup
5) Data Folder: A folder structure for data management
6) PySpark: Configuration for working with PySpark
7) Kedro-Viz: Kedro's native visualisation tool

Which tools would you like to include in your project? [1-7/1,3/all/none]:
 (none): 1-5
```

It will ask if you want an example pipeline. This one is optional, but maybe it is better to choose yes, to test if the installation works:

```
Would you like to include an example pipeline? :
 (no): yes

Congratulations!
Your project 'mlops_example' has been created in the directory
C:\Users\rosan\OneDrive\Desktop\trimester_2024\02_modular_code\mlops-example

You have selected the following project tools: ['Linting', 'Testing', 'Custom Logging', 'Do
cumentation', 'Data Structure']
It has been created with an example pipeline.

To skip the interactive flow you can run `kedro new` with
kedro new --name=<your-project-name> --tools=<your-project-tools> --example=<yes/no>
```

Now enter in the folder of the project, in my case it is *mlops-example* and run the following line:

```
cd mlops-example

uv pip install -r requirements.txt

uv pip install kedro-datasets

uv pip install kedro-viz

uv pip install pytest

uv pip install pytest-cov
```

Now you can install the plugin to use MLflow just by writing in the command line:

```
kedro mlflow init
```

```
 kedro mlflow init
As an open-source project, we collect usage analytics.
We cannot see nor store information contained in a Kedro project.
You can find out more by reading our privacy notice:
https://github.com/kedro-org/kedro-plugins/tree/main/kedro-telemetry#privacy-notice
Do you opt into usage analytics?  [y/N]: n
You have opted out of product usage analytics, so none will be collected.
'conf/local/mlflow.yml' successfully updated.
```

This new yml file in that location is the one that you should fill with the MLflow information.

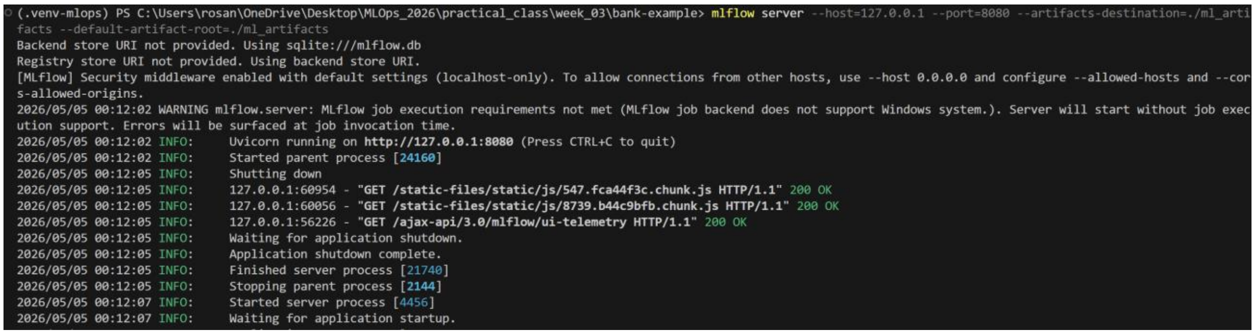
Now, if you install the start example before, make a kedro run by typing in the terminal

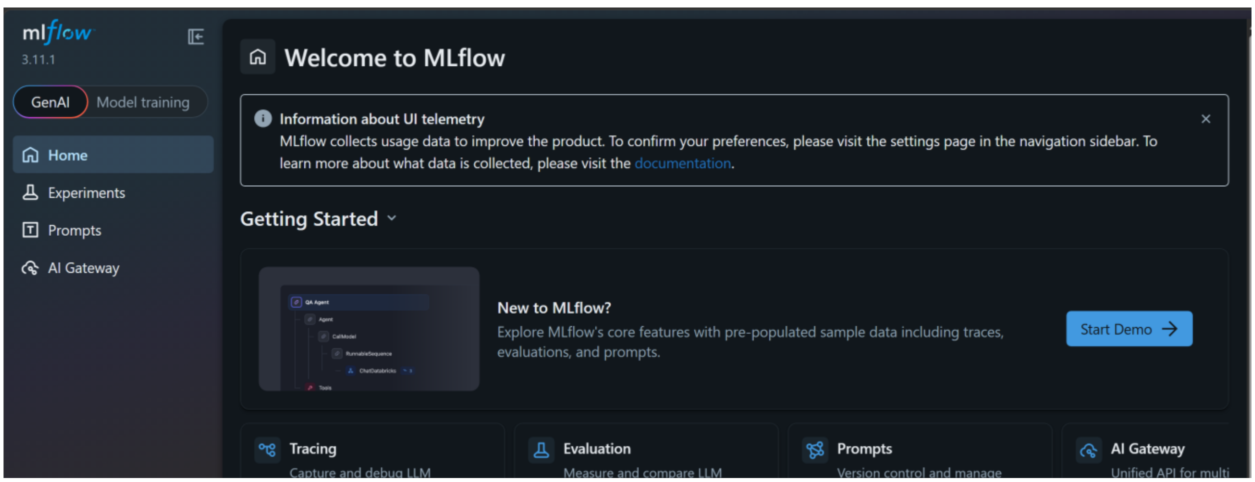
```
kedro run
```

```
(test_env) PS C:\Users\rosan\OneDrive\Desktop\trimester_2024\02_modular_code\mlops-example>
 kedro run
[05/06/24 23:25:28] INFO     Kedro project mlops-example                    session.py:324
                    INFO     Registering new custom resolver:            mlflow_hook.py:65
                             'km.random_name'
                    INFO     The 'tracking_uri' key in          kedro_mlflow_config.py:260
                             mlflow.yml is relative
                             ('server.mlflow_(tracking|registry)
                             _uri = mlruns'). It is converted to
                             a valid uri:
                             'file:///C:/Users/rosan/OneDrive/De
                             sktop/trimester_2024/02_modular_cod
                             e/mlops-example/mlruns'
2024/05/06 23:25:28 INFO mlflow.tracking.fluent: Experiment with name 'mlops_example' does not exist. Creating a new experiment.
[05/06/24 23:25:29] INFO     Using synchronous mode for loading and sequential_runner.py:64
                             saving data. Use the --async flag for
                             potential performance gains.
                             https://docs.kedro.org/en/stable/nodes
                             _and_pipelines/run_a_pipeline.html#loa
                             d-and-save-asynchronously
                                  file://C:\Users\rosan\anaconda3\envs\test_env\lib\site-
                                  packages\kedro\runner\sequential_runner.py (ctrl + click)
                    INFO     Loading data from companies            data_catalog.py:483
                             (CSVDataset)...
                    INFO     Running node: preprocess_companies_node:       node.py:361
                             preprocess_companies([companies]) ->
                             [preprocessed_companies]
[05/06/24 23:25:30] INFO     Saving data to preprocessed_companies  data_catalog.py:525
                             (ParquetDataset)...
                    INFO     Completed 1 out of 6 tasks         sequential_runner.py:90
                    INFO     Loading data from shuttles             data_catalog.py:483
                             (ExcelDataset)...
[05/06/24 23:25:33] INFO     Running node: preprocess_shuttles_node:        node.py:361
                             preprocess_shuttles([shuttles]) ->
                             [preprocessed_shuttles]
                    INFO     Saving data to preprocessed_shuttles   data_catalog.py:525
                             (ParquetDataset)...
                    INFO     Completed 2 out of 6 tasks         sequential_runner.py:90
                    INFO     Loading data from preprocessed_shuttles  data_catalog.py:483
                             (ParquetDataset)...
                    INFO     Loading data from preprocessed_companies data_catalog.py:483
                             (ParquetDataset)...
                    INFO     Loading data from reviews (CSVDataset)... data_catalog.py:483
                    INFO     Running node: create_model_input_table_node:   node.py:361
                             create_model_input_table([preprocessed_shuttles;pr
                             eprocessed_companies;reviews]) ->
                             [model_input_table]
                    INFO     Saving data to model_input_table       data_catalog.py:525
                             (ParquetDataset)...
                    INFO     Completed 3 out of 6 tasks         sequential_runner.py:90
                    INFO     Loading data from model_input_table    data_catalog.py:483
                             (ParquetDataset)...
                    INFO     Loading data from params:model_options data_catalog.py:483
                             (MemoryDataset)...
                    INFO     Running node: split_data_node:                 node.py:361
                             split_data([model_input_table;params:model_options
                             ]) -> [X_train;X_test;y_train;y_test]
                    INFO     Saving data to X_train (MemoryDataset)... data_catalog.py:525
                    INFO     Saving data to X_test (MemoryDataset)...  data_catalog.py:525
                    INFO     Running node: train_model_node:                node.py:361                    train_model([X_train;y_train]) -> [regressor]
```

```
                    INFO     Running node: train_model_node:                node.py:361          train_model([X_train;y_train]) -> [regressor]
[05/06/24 23:25:34] INFO     Saving data to regressor               data_catalog.py:525          (PickleDataset)...
                    INFO     Completed 5 out of 6 tasks         sequential_runner.py:90    INFO     Loading data from regressor         data_catalog.py:483          (PickleDataset)...
                    INFO     Loading data from X_test               data_catalog.py:483          (MemoryDataset)...
                    INFO     Loading data from y_test               data_catalog.py:483          (MemoryDataset)...
                    INFO     Running node: evaluate_model_node:             node.py:361          evaluate_model([regressor;X_test;y_test]) -> None
                    INFO     Model has a coefficient R^2 of 0.402 on test data. nodes.py:55    INFO     Completed 6 out of 6 tasks         sequential_runner.py:90          INFO     Pipeline execution completed su
ssfully.      runner.py:119(test_env) PS C:\Users\rosan\OneDrive\Desktop\trimester_2024\02_modular_code\mlops-example>
```

Now since we did not change anything in the configuration file, so the run of MLflow took place in the standard 5000 port. If you write mlfow ui (or the command from the practical classes mlflow, **mlflow server --host=127.0.0.1 --port=8080 --artifacts-destination=./ml_artifacts --default-artifact-root=./ml_artifacts**) in terminal, you will see:





Note: the command mlfow ui is just the standard one, with the standard port 5000. We still prefer to specify where the port should be opened and the folder of our ml artifacts.